

Efficient Large-Scale Expensive Optimization via Surrogate-Assisted Subproblem Selection

Kaili Zhao, Xilu Wang[✉], Associate Member, IEEE, Chaoli Sun[✉], Senior Member, IEEE, Yaochu Jin[✉], Fellow, IEEE, and Asad Hayat[✉]

Abstract—Traditional large-scale evolutionary algorithms are limited in their ability to solve certain real-world applications with high-dimensional, closed-box, and computationally expensive objectives due to their need for numerous objective evaluations. Surrogate-assisted evolutionary algorithms (SAEAs) have shown effective for expensive closed-box optimization by relying on inexpensive surrogate models. However, large-scale optimization remains challenging for SAEAs due to the exponentially growing search space and the presence of multiple local optima, resulting in difficulty in training a proper model due to the lack of samples. To address these challenges, we propose constructing an initial surrogate model on randomly selected dimensions and calculating a Gaussian distribution for each sampled dimension. The surrogate then provides predictions when perturbing each sampled dimension by sampling from the distribution, enabling the identification of the most important variables for constructing an active subproblem to reduce the search space. A secondary surrogate model, built for the active subproblem, guides the offspring generation and environmental selection for a modified particle swarm optimization algorithm to effectively explore the subspace while escaping local optima in large-scale problems. Experimental results on CEC'2013 and CEC'2010 benchmark problems show that the proposed method outperforms state-of-the-art algorithms in addressing large-scale expensive optimization problems. The efficiency of the proposed method is further verified on CEC'2010 benchmark problems extended to 2000 dimensions.

Index Terms—Large-scale expensive optimization problems (LSEOPs), radial basis function (RBF), social learning particle swarm optimization (SL-PSO), surrogate-assisted optimization, surrogate-assisted subproblem selection.

Received 12 August 2024; revised 22 November 2024 and 5 January 2025; accepted 14 February 2025. Date of publication 20 February 2025; date of current version 8 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62372319, and in part by the Key Research and Development Program Project of Shanxi Province under Grant 202102020101002. This article was approved by Associate Editor R. Cheng. (Corresponding author: Chaoli Sun.)

Kaili Zhao is with the School of Electronic Information Engineering, Taiyuan University of Science and Technology, Taiyuan 030024, China, and also with the Department of Mechanical Engineering, Shanxi Engineering Vocational College, Taiyuan 030009, China (e-mail: kaili.zhao@stu.tyust.edu.cn).

Xilu Wang is with the School of Computer Science and Electronic Engineering, University of Surrey, GU2 7XH Surrey, U.K. (e-mail: wangxilu@surrey.ac.uk).

Chaoli Sun is with the School of Computer Science and Technology, Taiyuan University of Science and Technology, Taiyuan 030024, China (e-mail: chaoli.sun@tyust.edu.cn).

Yaochu Jin is with the School of Engineering, Westlake University, Hangzhou 310030, China (e-mail: jinyaochu@westlake.edu.cn).

Asad Hayat is with the School of International Education, Taiyuan University of Science and Technology, Taiyuan 030024, China (e-mail: asadhat@stu.tyust.edu.cn).

Digital Object Identifier 10.1109/TEVC.2025.3544449

I. INTRODUCTION

OPTIMIZATION problems involving a substantial number of variables are frequently encountered in real-world applications across various domains. Examples include job shop scheduling [1], casting sequence scheduling [2], integer linear programming [3], and air route network optimization [4]. In general, a minimization problem can be formally expressed as follows:

$$\min f(\mathbf{x}) \quad (1)$$

$$\text{s.t. } \mathbf{x} \in \Omega \subseteq \mathbb{R}^D \quad (2)$$

where $\Omega \subseteq \mathbb{R}^D$ represents the search space, $\mathbf{x} = (x_1, \dots, x_D)^T$ denotes the decision vector, and D is the number of decision variables. A problem is considered a large-scale optimization problem (LSOP) when $D \geq 100$ [5], [6]. Furthermore, a problem is termed a closed-box problem if the objective function $f(\mathbf{x})$ cannot be explicitly formulated or if its analytical properties (such as gradients or Hessians) are unknown or unavailable [7].

In the past decades, numerous evolutionary algorithms (EAs), without requiring explicit expression and differentiation of the objective function, have been developed to address LSOPs, which are called large-scale EAs (LSEAs) [8], [9]. LSEAs can be categorized into three groups: 1) divide-and-conquer-based LSEAs; 2) dimension reduction-based LSEAs; and 3) novel search strategy-based LSEAs. In the first category, the “divide-and-conquer” strategy in evolutionary optimization addresses high-dimensional problems by decomposing them into multiple lower dimensional subproblems, each focusing on a subset of the original variables. Then, each subproblem is optimized using an EA, and the optimal solutions found for the subproblems are combined to construct a solution for the original high-dimensional problem [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20].

As a result, the performance of such a strategy generally heavily relies on the construction of the subproblems. The second group of approaches focuses on dimensionality reduction, transforming the original large-scale problem into a medium-to-low-scale problem. The commonly used dimension reduction techniques include principal component analysis (PCA) [21], problem transformation [22], mapping [23], and so on.

In the third category, new optimization strategies are proposed to generate solutions, aiming to improve the diversity of population and better balance exploration and exploitation

for LSOPs with more than 1 000 dimensions [24], [25], [26], [27], [28].

Despite the success of large-scale EAs, they often require numerous objective evaluations to locate the optimal or near-optimal solutions due to the exponentially growing search space or the complex fitness landscape. However, some real-world applications, such as the optimization of carbon fiber composite materials [29], [30], are not only large-scale but also computationally expensive to evaluate, making it impossible to allow numerous objective evaluations. Numerous studies have suggested that surrogate models are one of the most promising ideas to reduce the expensive evaluations [31], [32], [33]. Surrogate models are commonly used to replace computationally expensive objective functions to assist in the optimization of expensive problems [34], [35], [36]. Commonly used surrogate models include support vector machines (SVMs) [37], polynomial regression (PR) [38], Gaussian process (GP) [39], artificial neural networks (ANNs) [32], and radial basis function (RBF) networks [40]. For leveraging the surrogate model to address expensive LSOPs, several surrogate-assisted large-scale optimization algorithms have been proposed.

Surrogate-assisted large-scale optimization algorithms can be classified into two categories. Methods in the first category utilize the approximated values provided by the surrogate model solely for environmental selection. Tian et al. [39] used a GP model to simultaneously provide the approximated objective value and approximation uncertainty for each solution. Some solutions were then selected at each generation of optimization for real objective evaluation according to the nondominance relationship between the approximated value and approximation uncertainty for each generation. Cheng et al. [41] put forward a multisurrogate-assisted stochastic fractal search algorithm in which RBF and GP models are used in the first and second stages, respectively. The solution with the minimum approximated value in the first stage and that with the maximum expected improvement (EI) value in the second stage were selected for real expensive evaluation.

Methods in the second category for high-dimensional expensive problems regard the surrogate models as the optimization objective to guide the search for the optimal solution of the expensive optimization problem. For example, Yu et al. [42] utilized an RBF-assisted social learning particle swarm optimization (RBF-assisted SL-PSO) to search for a good solution and provide for particle swarm optimization to learn. Wang et al. [43] proposed an evolutionary sampling assisted optimization algorithm in which a global exploration and local exploitation are alternatively conducted. The global RBF model was used for prescreening the best solution in the offspring and the search was conducted for finding the optimum of the local model to participate in the global search. However, it is to be noted that most surrogate-assisted EAs (SAEAs) are only able to solve large-scale expensive optimization problems that do not exceed 200 dimensions, which are also called high-dimensional expensive optimization problems [44].

The literature review shows that the “divide-and-conquer” technique is quite efficient for solving large-scale expensive optimization problems (LSEOPs) [45], [46]. Specifically, the “divide-and-conquer” technique is particularly well-suited for parallel computing frameworks [47], [48], [49]. By searching for optimal solutions in parallel, the optimization process can be significantly accelerated, leveraging hardware capabilities rather than relying solely on algorithmic efficiency. To further reduce optimization time from an algorithmic perspective, we propose solving a subproblem that consists of a smaller subset of variables from the original large-scale problem. In the proposed approach, a subproblem is formed by randomly selecting variables from the original problem at first.

Then, a surrogate model is trained for the refined subproblem and regarded as an optimization objective for guiding the exploration of the large-scale expensive problem. The main contributions of this article are given as follows.

- 1) An active subproblem of LSEOP, instead of a number of subproblems, is evolved at each generation to save the optimization time. A surrogate model is built on dimensions randomly sampled from LSEOP, helping identify critical dimensions to form a smaller, active subproblem. A second model is then trained on this active subproblem to facilitate environmental selection, thereby accelerating the search for the optimal solution of the LSEOP.
- 2) A modified SL-PSO is proposed for generating offspring for the active subproblem. In the proposed approach, the velocity of each solution is updated by learning from a demonstrator on the active subproblem to explore better solutions for the LSEOP, while also considering the best position found so far for the LSEOP to accelerate convergence.
- 3) In addition to conducting experiments on the CEC’2013 benchmark, we first extend the dimensionality to 2000 and test our method on the CEC’2010 benchmark. The experimental results indicate that our proposed method maintains its superior performance across higher dimensions.

The remainder of this article is organized as follows. Section II introduces the background of this work. Section III provides a comprehensive and detailed description of the proposed algorithm. In Section IV, the experimental results are given, and some analyses are presented. Finally, Section V concludes the article and discusses future work.

II. BACKGROUND

A. SAEAs for LSEOPs

To the best of our knowledge, the application of SAEAs for addressing large-scale expensive optimization problems has been effective for problems of moderate dimensionality, typically up to approximately 200 dimensions [44], [50]. Their efficacy for problems with 1000 or more dimensions remains highly limited.

This limitation stems from the fundamental challenge in surrogate modeling: as the dimensionality of the problem increases, more training data are required to construct an

accurate surrogate model. For computationally expensive problems, obtaining sufficient training data becomes increasingly challenging. To address this issue, researchers have developed strategies to decompose large-scale problems into a set of subproblems. These approaches aim to reduce the dimensionality of individual subproblems, thereby lowering the data requirements for surrogate model training. The subproblems are then cooperatively optimized with the assistance of surrogate models. For example, De Falco et al. [45], [46] introduced a surrogate-assisted cooperative co-evolutionary (SACC) algorithm in which an LSOP was divided into a number of subproblems and a surrogate model was trained for each subproblem to guide the search for its optimal solution. Fu et al. [51] employed the random feature selection technique to form a set of subproblems and updated the population of the large-scale problem by sequentially optimizing subproblems. Sun et al. [52] proposed to divide an LSOP into a number of low-dimensional subproblems using the random grouping technique, and a surrogate ensemble is used to assist the optimization of each subproblem. Recently, Gu et al. [53] constructed multiple subspaces based on the PCA and random decision variable selections for large-scale expensive problems and proposed to use three strategies to search for optimal solutions. Si et al. [54] proposed to train surrogate models on a number of linear subspaces, which are constructed by a pair of solutions generated from the set of elite solutions, instead of the original decision space.

B. Radial Basis Function Network

RBF networks (RBFN) are commonly used as surrogate models due to their scalability and efficiency for approximating expensive objective functions.

RBFN is a feedforward neural network consisting of three layers: 1) an input layer; 2) a hidden layer; and 3) an output layer. The hidden layer applies a RBF to the input. The commonly used RBFs include linear, cubic, thin plate spline, multiquadric, and Gaussian. In this article, the cubic function is adopted. The output of an RBF network is a linear combination of RBFs, given by

$$\hat{f}(\mathbf{x}) = \sum_{k=1}^{nc} \omega_k \varphi(\mathbf{x}, \mathbf{c}_k) + \omega_0 \quad (3)$$

where

$$\varphi(\mathbf{x}, \mathbf{c}_k) = \phi(\|\mathbf{x} - \mathbf{c}_k\|). \quad (4)$$

In (3) and (4), nc represents the number of nodes in the hidden layer, $\mathbf{c}_k$ is the center of the k th hidden node. ω_k is the k th coefficient, and ω_0 is a bias term. It has been shown in [55] that the performance of RBFN is less sensitive to the dimension of the problem to a certain extent. Thus, it is often employed as a surrogate model for solving large-scale problems.

III. PROPOSED ALGORITHM

This section presents the efficient large-scale expensive optimization via surrogate-assisted subproblem selection, denoted as LSEO-S3. Initially, the overall framework of the proposed algorithm is introduced, followed by a detailed

Algorithm 1: Pseudocode of LSEO-S3

Input: N_{Arc} : the number of initially sampled solutions;
 N : the population size.

Output: $\mathbf{x}_{gbest}, f(\mathbf{x}_{gbest})$

- 1 **Initializing a database:** Generate N_{Arc} solutions using the Latin hypercube sampling (LHS) method, evaluate them through the real expensive objective function, and store them in Arc ;
- 2 **Initializing a population pop :** Initialize N individuals to be the parent population (pop) for optimization, along with initializing the velocity of each individual;
- 3 **while the terminal criterion is not met do**
- 4 Select a number of important dimensions from the original large-scale problem to create an active subproblem SP^{dr} through the proposed surrogate-assisted subproblem selection strategy (**Details refer to Algorithm 2**);
- 5 Generate an offspring O^{dr} using the RBF-assisted SL-PSO variant for SP^{dr} (**Details refer to Algorithm 3**);
- 6 Update the subpopulation by implementing surrogate-assisted environmental selection (**Details refer to Section III-D**);
- 7 Replace the population pop on corresponding dimensions with the updated subpopulation;
- 8 Utilize the optimal solution from the subproblem to replace the corresponding dimensions of the global best solution $\mathbf{x}_{gbest}$, thereby generating a new solution $\mathbf{x}_{new}$;
- 9 Evaluate the $\mathbf{x}_{new}$ using the real expensive objective function, store it in the Arc ;
- 10 **end**
- 11 **Output** $\mathbf{x}_{gbest}$ and its corresponding $f(\mathbf{x}_{gbest})$ in the Arc .

description of the surrogate-assisted subproblem selection and the corresponding single-iteration evolve strategy. Finally, the section concludes with a complexity analysis, providing theoretical justification for the efficiency of LSEO-S3 in addressing large-scale expensive optimization problems.

A. Overall Framework

Previous research has demonstrated that decomposing an LSOP into a set of subproblems is a promising approach. Typically, these subproblems are constructed using random grouping or feature selection techniques. The optimal solution is then obtained through cooperative optimization of each subproblem, assisted by surrogate models. Despite numerous studies in this direction, the research community widely acknowledges two critical challenges: 1) the construction and 2) the optimization of subproblems [51], [52], [53], [54]. First, the method of constructing subproblems plays a crucial role in large-scale optimization, as it directly impacts the quality and efficiency of the overall optimization process. Second, taking no account of leveraging hardware capabilities, optimizing all subproblems before conducting real objective

evaluations at each iteration is computationally expensive and time-consuming.

To address these challenges, we propose an effective SAEA for forming an active subproblem that capture the most influential variables. This approach aims to focus computational resources on the most impactful variables of the optimization problem. Furthermore, to enhance efficiency, we develop a single-iteration evolve method based on SL-PSO for the active subproblem. This strategy significantly reduces computational cost while maintaining optimization quality. The details of the proposed LSEO-S3 method are presented in Algorithm 1, from which we can see that there are five main steps for LSEO-S3: 1) initialization; 2) surrogate-assisted dimension reduction; 3) generation of offspring for the subproblem; 4) surrogate-assisted environment selection; and 5) infill sampling.

- 1) *Initialization*: As shown in Algorithm 1, the initialization process includes the initialization of training samples and the population. For training samples, we generate N_{Arc} solutions using latin hypercube sampling (LHS), evaluate them with expensive objective functions, and store the results in the archive (Arc). Meanwhile, we create an initial population pop of N solutions using LHS without objective function evaluation, and initialize their velocities to leverage the SL-PSO algorithm.
- 2) *Surrogate-Assisted Dimension Reduction*: Surrogate-assisted dimension reduction aims to select some dimensions from the original large-scale problem to form an active subproblem for later updates. To achieve this, D_r dimensions are randomly selected from the LSOP to create an initial subproblem (SP^r) at first. The optimal solution $\mathbf{x}_{gbest}$ found so far is down-dimensionalized based on the dimensions of the initial subproblem (SP^r) to construct the optimal solution $\mathbf{x}_{best}^{SP^r}$ for the subproblem. A surrogate model is then trained for SP^r , which approximates the objective value of the optimal solution of this subproblem ($\mathbf{x}_{best}^{SP^r}$) and those variant that only one dimension is different to the optimal solution. In the second step, variables with great impacts on the objective values are identified to form an active subproblem (SP^{dr}) for this round. The details of surrogate-assisted dimension reduction are given in Section III-B.
- 3) *Offspring Generation for the Subproblem*: After the dimension of the active subproblem is determined, a new RBF model is trained for the subproblem. An offspring population is then generated using the proposed SL-PSO variant. Note that, instead of running an EA for several iterations for the subproblems, we only generate the offspring population for one iteration, which can further reduce the computational time.
- 4) *Surrogate-Assisted Environmental Selection*: The RBF model is then used to approximate each solution in the parent and offspring populations on the subproblem, based on which a new parent population for the subproblem is selected. The current population for subproblem SP^{dr} will replace the population pop on corresponding

dimensions to update pop . A simple example of population updating is presented in Fig. 1.

- 5) *Infill Sampling*: Additionally, the best solution found so far for the subproblem SP^{dr} will substitute the solution with the best objective value in the archive Arc to form a new solution, which will be evaluated using the real objective function and saved to the archive Arc . If the maximum number of objective evaluations is reached, the optimal solution in the Arc will be output.

In the following, the surrogate-assisted dimension reduction, offspring population generation, surrogate-assisted environmental selection, infill sampling and complexity analysis are presented in detail.

B. Surrogate-Assisted SubProblem Selection

Research across various problem domains has consistently revealed that not all dimensions equally influence the objective function. This phenomenon, often referred to as “diminished effective dimension” [56], suggests that the objective function only relies on a subset of dimensions, which shed lights on constructing low-dimensional subproblems for the large-scale problems. However, identifying important dimensions within the constraints of a limited evaluation budget is a challenging task when solving LSEOPs.

In our proposed LSEO-S3 algorithm, some decision variables of the LSOP will be randomly selected to form an initial subproblem (SP^r). To further identify the important variables for the optimization and create an active subproblem with fewer representative dimensions, we propose to employ a surrogate model for feature selection. The pseudocode of the proposed method for the formation of a representative subproblem is given in Algorithm 2.

In the surrogate-assisted dimension reduction method, a surrogate model M^r is trained for the initial subproblem SP^r using $N_s = 3 \times D_r$ samples randomly selected from Arc . Using the predictions of M^r , we analyze the optimal solution $\mathbf{x}_{best}^{SP^r} = (x_{best,1}^{SP^r}, x_{best,2}^{SP^r}, \dots, x_{best,D_r}^{SP^r})$ for SP^r by systematically perturbing each dimension to observe its influence on the objective function. This process involves carefully modifying each dimension of the optimal solution and using M^r to predict the resulting changes in objective values.

To achieve this, a Gaussian distribution for each variable j is estimated by calculating the mean and variance of the values of the corresponding dimension among the subpopulation P^r

$$\mathbf{x}_j \sim \mathcal{N}(\mu_j, \sigma_j^2) \quad (5)$$

After obtaining the Gaussian distribution of each variable in the selected dimensions D_r , the perturbation can be designed as sampling from the Gaussian distribution to replace the corresponding value in the optimal solution $\mathbf{x}_{best}^{SP^r}$ one by one. The objective value of the optimal solution after the perturbation is predicted by the surrogate model M_r . Hence, the change in objective values between the optimal point and its perturbation on the j th variable can be calculated as follows:

$$\text{diff}_j = |\hat{f}(\mathbf{x}_{best}^{SP^r, j}) - \hat{f}(\mathbf{x}_{best}^{SP^r})|. \quad (6)$$

After obtaining the influence of each variable on the objective function indicated by the difference in (6), the mean of all the

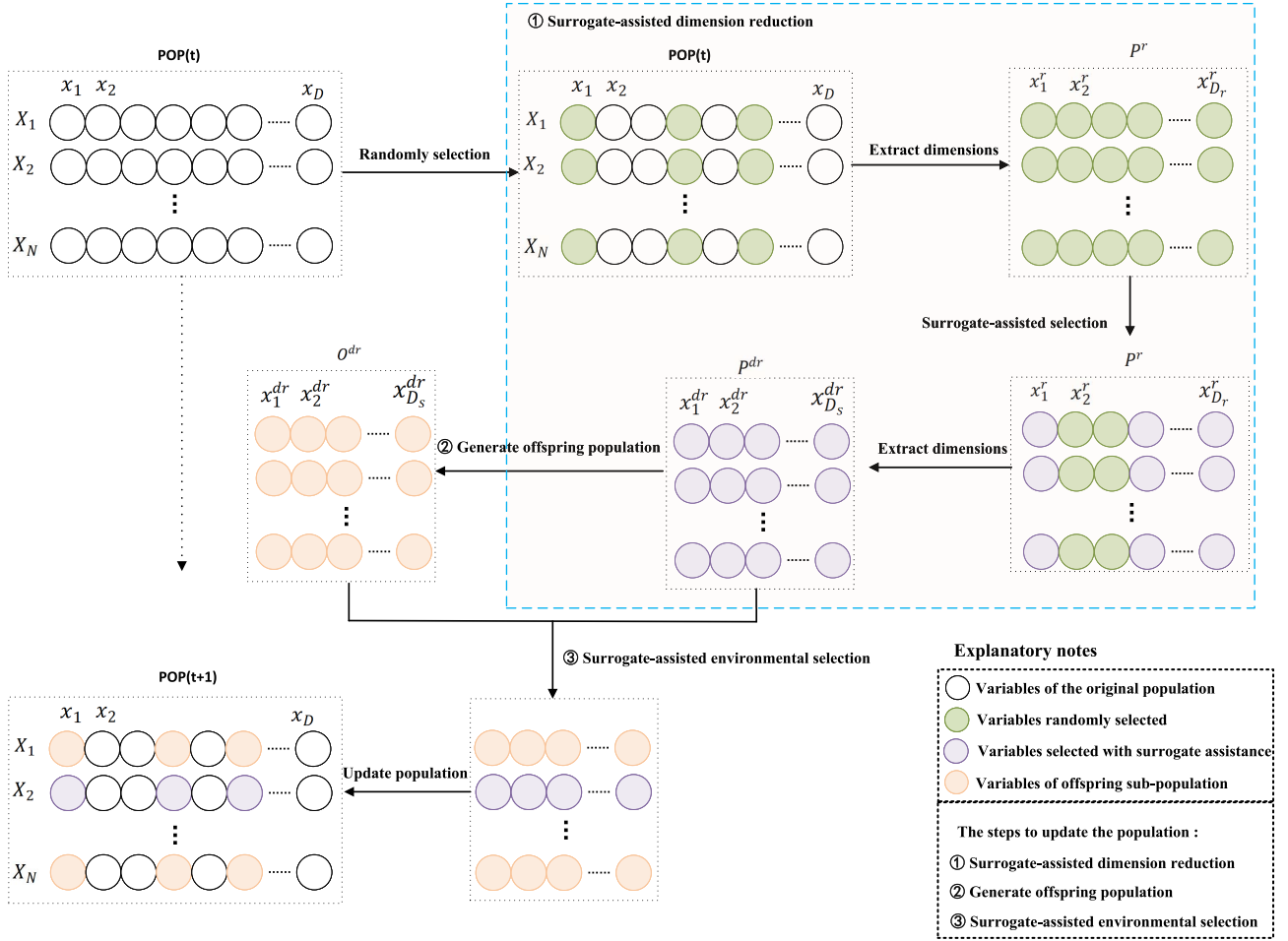


Fig. 1. Schematic of population updating.

differences is calculated and serves as a threshold. As a result, the variables having larger differences than the threshold are selected to construct the final subproblem for optimization in the current round.

By examining how these changes affect the objective function near the optimal point, we can identify the most sensitive dimensions—those that have the greatest impact on solution quality. This approach allows us to focus on the most promising areas of the search space, effectively reducing the dimensionality of the problem and enhancing the efficiency of the optimization process.

C. Surrogate-Assisted Reproduction for SubProblem

After the subproblem is finally determined, a modified SL-PSO reproduction formula assisted by an RBF surrogate model is proposed to generate offspring for subproblem. The SL-PSO algorithm has a good performance in solving LSOPs due to its strong exploration capability. However, the speed in converging to the global optimal position is much lower. Thus, we propose a modified dynamic to update the velocity of solutions for SL-PSO to leverage its good exploration performance and improve the convergence speed as well, as shown in the following:

$$v_{i,j}(t+1) = r_1(t) \cdot v_{i,j}(t) + r_2(t) \cdot I_{i,j}(t) + r_3(t) \cdot G_{i,j}(t) \quad (7)$$

with

$$\begin{cases} I_{i,j}(t) = x_{k,j}(t) - x_{i,j}(t) \\ G_{i,j}(t) = x_{\text{best},j}(t) - x_{i,j}(t). \end{cases} \quad (8)$$

The position is updated as follows:

$$x_{i,j}(t+1) = x_{i,j}(t) + v_{i,j}(t+1). \quad (9)$$

In (7)–(9), $v_{i,j}(t)$ and $x_{i,j}(t)$ denote the velocity and position of solution i on j th dimension, respectively. $x_{k,j}$ represents the j th dimension of the demonstrator solution k which has a better estimated value than solution i from RBF model. $x_{\text{best},j}$ denotes the j th dimension of the best solution found so far, $r_1(t)$, $r_2(t)$, and $r_3(t)$ are three numbers generated uniformly in the range of $[0, 1]$ at t th generation.

From (7), we can see that the update of $v_{i,j}(t+1)$ comprises three components. The first component represents the inertia term, which maintains the current direction of movement for the solution. The second component signifies the velocity adjustment for the solution on any dimension by learning from a demonstrator with a better approximated value on the active subproblem, encouraging the exploration of new regions for potential high-quality solutions. The third component directs the solution toward the best position found so far, accelerating convergence toward the global optimum. The pseudocode of

Algorithm 2: Dimension Reduction Assisted by Surrogate Model

Input: pop : current population; D_r : size of initial subproblem; Arc : database; $\mathbf{x}_{gbest}$: the best solution in Arc found so far.

Output: SP^{dr} : the formed subproblem after dimension reduction at current round.

- 1 Randomly select D_r variables from the large-scale problem to form an initial subproblem SP^r ;
- 2 Randomly select $N_s = 3 \times D_r$ solutions from Arc and train an RBF model (M^r) for SP^r ;
- 3 Copy the position from $\mathbf{x}_{gbest}$ on corresponding dimensions of SP^r , denoted as $\mathbf{x}_{best}^{SP^r}$;
- 4 Approximate the objective value of $\mathbf{x}_{best}^{SP^r}$, denoted as $\hat{f}(\mathbf{x}_{best}^{SP^r})$;
- 5 Copy position of each solution from population pop on corresponding dimensions to form a subpopulation P^r ;
- 6 **for** $j = 1$ **to** D_r **do**
- 7 Calculate the mean $\mu(j)$ and variance $\sigma(j)$ of the j th dimension data of the population P^r , and perform Gaussian sampling, referred as $x_{GP,j}^{SP^r}$;
- 8 Replace $x_{best,j}^{SP^r}$ using $x_{GP,j}^{SP^r}$ and generate a new solution ($\mathbf{x}_{best}^{SP^r,j}$);
- 9 Approximate the objective value of $\mathbf{x}_{best}^{SP^r,j}$, denoted as $\hat{f}(\mathbf{x}_{best}^{SP^r,j})$;
- 10 $diff_j = |\hat{f}(\mathbf{x}_{best}^{SP^r,j}) - \hat{f}(\mathbf{x}_{best}^{SP^r})|$;
- 11 **end**
- 12 $Mdiff = \frac{\sum_{j=1}^{D_r} diff_j}{D_r}$;
- 13 The variables satisfying $diff_j \geq Mdiff$, $j = 1, 2, \dots, D_r$ will form a new subproblem, denoted as SP^{dr} at the current round.

surrogate-assisted reproduction for the subproblem is given in Algorithm 3.

D. Surrogate-Assisted Environmental Selection

Canonical SL-PSO lacks an explicit environmental selection process, which leads to its slow convergence and inability to find near-optimal solutions within limited computational budgets. To address this, we incorporate environmental selection as follows. At first, the offspring population O^{dr} is combined with the parent populations P^{dr} . Then, the objective value of each solution in the combined population is predicted by the model M^{dr} and sorted in ascending order. Finally, the first N solutions are selected to update the positions of the population for large-scale optimization on corresponding dimensions.

E. Infill Sampling

In SAEAs, the infill sampling strategy plays a crucial role in finding good solutions within a limited computational budget. In our method, the best solution found so far for each subproblem is used to replace the solution with minimum objective value in archive Arc on corresponding dimensions to generate a new solution. The new solution is evaluated using

Algorithm 3: Surrogate-Assisted Reproduction for SubProblem

Input: pop : current population; N : population size; $\mathbf{x}_{gbest}$: the best solution in Arc found so far; D_s : the dimension of subproblem SP^{dr} .

Output: The offspring of subproblem O^{dr} .

- 1 Train a surrogate model M^{dr} for the subproblem SP^{dr} ;
- 2 Copy position and velocity of each solution from population pop on corresponding dimensions to form an initial population P^{dr} ;
- 3 Approximate the objective value for each solution in P^{dr} using M^{dr} ;
- 4 Copy the position from $\mathbf{x}_{gbest}$ on corresponding dimensions of SP^{dr} , denoted as $\mathbf{x}_{best}$;
- 5 **for** $i = 1$ **to** N **do**
- 6 Choose a solution $\mathbf{x}_k$ with better approximated value than $\mathbf{x}_i$ from the population P^{dr} ;
- 7 **for** $j = 1$ **to** D_s **do**
- 8 Update velocity and position for solution i on j th dimension using Eqs. (7) and (9), respectively;
- 9 **end**
- 10 **end**
- 11 Output the offspring population O^{dr} .

a real objective function and saved to the archive. The fitness value of this new solution is then compared to that of $\mathbf{x}_{gbest}$. If the new solution proves superior, $\mathbf{x}_{gbest}$ is updated accordingly.

F. Complexity Analysis

1) *Complexity of the Proposed Algorithm:* The computational complexity of the proposed LSEO-S3 primarily stems from three main components: 1) surrogate-assisted dimension reduction; 2) offspring population generation; and 3) surrogate-assisted environmental selection. Other operations, such as initialization and infill sampling, can be neglected when evaluating the overall computational complexity.

Surrogate-assisted dimension reduction involves two steps: 1) random selection and 2) surrogate-assisted selection. The complexity of random selection is relatively low and can be neglected. The second step involves training an RBF model, which has a computational complexity of $O((N_r)^2 \times D_r)$ (N_r is the size of the training data and D_r is the dimension of RBF input). Additionally, the complexity of selecting important dimensions through Gaussian sampling is $O(D_r)$ (D_r is the dimension of the initial subproblem).

For the generation of the offspring population and surrogate-assisted environment selection after dimension reduction, an RBF model is trained to generate offspring and perform environment selection, with a computational complexity of $O((N_r)^2 \times D_s)$. Subsequently, N offspring are generated for the subproblem, which results in a complexity of $O(N \times D_s)$ (D_s is the number of dimensions after the dimension reduction process).

Therefore, we can see that the overall complexity of the proposed method mainly lies in the model building. Since this proposed method constructs only two RBF models, the overall complexity remains relatively low. The complexity of the proposed algorithm, denoted as $T_{LSEO-S3}$, is given by $O((N_r)^2 \times D_r + (N_r)^2 \times D_s)$.

2) *Complexity Comparison*: The complexities of algorithms SEA-LEEA-RG, SAEA-RFS, SADE-AMSS, and L2SMEA are compared based on the different types of surrogate models used in their training. In the following, N_r represents the size of the training data, k represents the number of subproblems in each generation, and d represents the dimension of a subproblem.

- 1) *SEA-LEEA-RG*: This algorithm generates about k subproblems per generation, and each subproblem trains two RBF models for integration, which assisted in the optimization process. The complexity of SEA-LEEA-RG is $T_{\text{SEA-LEEA-RG}} = O(2 \times k \times (N_r)^2 \times d)$ (where $k = 20$, and d is an integer randomly generated in the range $[1, 100]$).
- 2) *SAEA-RFS*: This algorithm generates k subproblems per generation, and each subproblem trains an RBF model to assist in the optimization process. The complexity of SAEA-RFS is $T_{\text{SAEA-RFS}} = O(k \times (N_r)^2 \times d)$ (where $k = 20$ and d is an integer randomly generated in the range $[1, 100]$).
- 3) *SADE-AMSS*: It is consistent with the SAEA-RFS. The complexity of SADE-AMSS is $T_{\text{SADE-AMSS}} = O(k \times (N_r)^2 \times d)$ (where $k = 20$ and d is an integer randomly generated in the range $[1, 100]$).
- 4) *L2SMEA*: This algorithm constructs eight subproblems per generation and utilizes the Kriging surrogate model to assist in the optimization process. Thus, the complexity of L2SMEA is $T_{\text{L2SMEA}} = O(k \times (N_r)^3)$ (where $k = 8$).

Based on the above analysis, as $T_{\text{LSEO-S3}} \leq O(2 \times D_r \times (N_r)^2)$, $T_{\text{SEA-LEEA-RG}} \leq O(40 \times D_r \times (N_r)^2)$, and $T_{\text{SAEA-RFS}} \leq O(20 \times D_r \times (N_r)^2)$, we can see that LSEO-S3 can gain more computational efficiency than SEA-LEEA-RG, SAEA-RFS, and SADE-AMSS. For L2SMEA, $T_{\text{L2SMEA}} \gg O((N_r)^2)$, its computational complexity is much larger than LSEO-S3.

Overall, we can see that our proposed LSEO-S3 algorithm can gain computational efficiency by the method of surrogate-assisted subproblem selection.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we first conduct ablation experiments to assess the impact of two key components of our approach, the surrogate-assisted dimension reduction technique and the modified SL-PSO algorithm. Then, we compare our method's performance on the CEC'2013 test problems against four state-of-the-art EAs designed for expensive LSOPs. Furthermore, we extend the dimensionality of CEC'2010 test problems to 2000 dimensions to verify the efficiency of our proposed method.

A. Benchmark Functions and Parameter Settings

The performance of our proposed algorithm is verified by conducting experiments on CEC'2013 benchmark functions [57]. The CEC'2013 benchmark functions consist of a total of 15 problems, which are categorized into five distinct categories: 1) fully separable functions (F1–F3); 2) partially additive separable functions with a separable subcomponent

(F4–F7); 3) partially additive separable functions with no separable subcomponents (F8–F11); 4) overlapping functions (F12–F14); and 5) nonseparable functions (F15). The test suite consists of 15 functions with diverse characteristics. Seven of these test instances (F1, F4, F8, F11, F13, F14, and F15) exhibit asymmetric single-peak landscapes, while the remaining functions feature asymmetric multi-peak landscapes. Most functions are 1000-D, with the exception of F13 and F14, which have 905 dimensions. The functions also vary in their variable bounds: F1, F5, and F9 have variables constrained within $[-5, 5]$; F3, F6, and F10 have bounds of $[-32, 32]$; and the remaining functions have variables bounded within $[-100, 100]$. This diverse set of test problems allows for a comprehensive evaluation of optimization algorithms across various landscape complexities, dimensionalities, and search space sizes.

The algorithm begins by randomly generating and evaluating 500 solutions using the expensive objective function. These solutions are then stored in an archive denoted as *Arc*. The main population size, N , is set to 100 individuals. Many studies [44], [58], [59] have demonstrated the efficiency of RBF for problems with up to 200 dimensions. Considering both accuracy and training time, we choose $D_r = 100$ as the dimension of the initial subproblem in our method. The upper and lower bounds of velocity are set to be 10% of those of positions of large-scale problems.

To ensure a fair comparison, the stopping criterion for all algorithms in experiments is set to reach the maximum number of expensive function evaluations, which is set to $11 \times D$. It is worth noting that when the dimension is 2000, the available number of expensive function evaluations is set to 10000. Additionally, each algorithm is executed independently for a total of 25 runs. The Wilcoxon rank sum test [60] with a significance level of 0.05 is used for the statistical comparison of the median values of the obtained optimal values on 25 independent runs. Symbols “+,” “ $\approx$,” and “-” in Tables I–V indicate that LSEO-S3 is significantly better, equivalent to, and worse than the compared algorithms, respectively.

B. Effectiveness of the Surrogate-Assisted Dimension Reduction

This section evaluates the effectiveness of the surrogate-assisted dimension reduction strategy by comparing LSEO-S3 with a variant called LSEO-S3-R. We conduct this assessment using a representative subset of CEC'2013 benchmark functions: F1, F7, F11, F14, and F15. These functions were selected to cover each category of the benchmark suite, ensuring a comprehensive evaluation. The key distinction in LSEO-S3-R is that it randomly selects D_r dimensions from the large-scale problem without applying any additional dimension reduction procedures.

The statistical results of LSEO-S3 and LSEO-S3-R are presented in Table I. From Table I, it can be seen that LSEO-S3 consistently achieves better performances on all five test problems, validating the efficiency of the proposed surrogate-assisted dimension reduction. A possible explanation for this observation lies in the method of subproblem

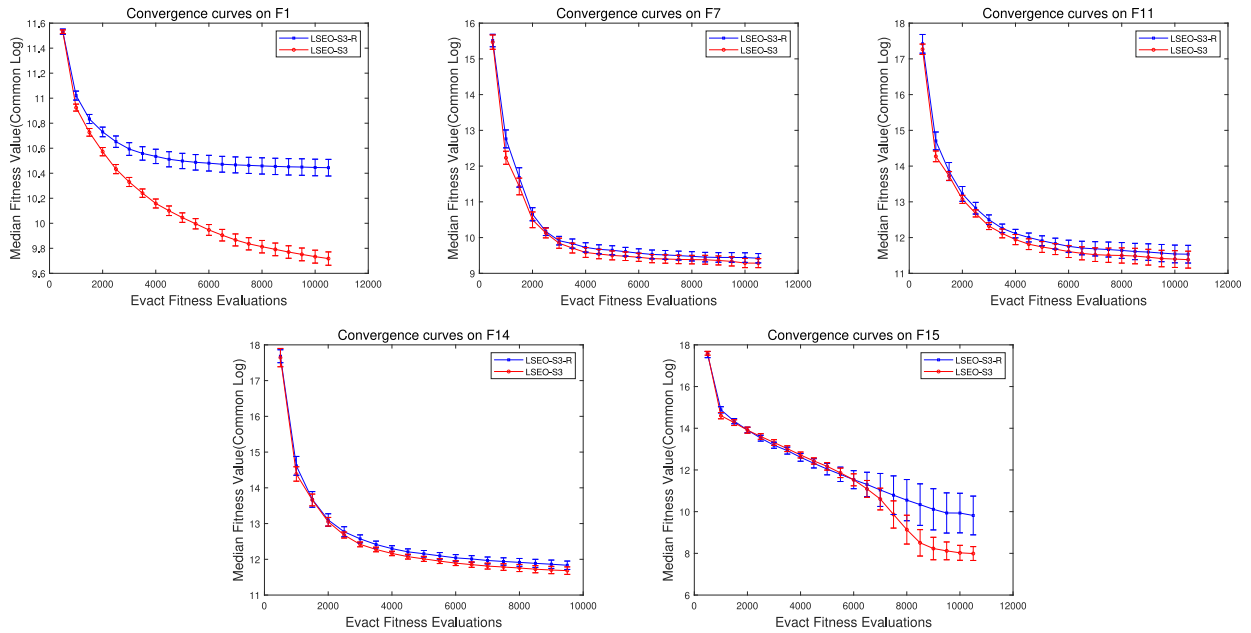


Fig. 2. Convergence trends comparison on F1, F7, F11, F14, F15. The vertical axis represents the median value of the optimal solution found so far on 25 independent runs, and the horizontal axis represents the number of objective evaluations.

TABLE I

MEDIAN AND MEDIAN ABSOLUTE DEVIATION (MAD) OBTAINED BY LSEO-S3-R AND LSEO-S3 ON F1, F7, F11, F14, F15 TEST PROBLEMS. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED

Prob.	LSEO-S3-R	LSEO-S3
F1	2.4786E+10(2.7437E+09) +	5.0344E+09(4.6611E+08)
F7	2.3903E+09(5.3693E+08) +	1.8950E+09(4.4321E+08)
F11	3.6836E+11(6.6065E+10) ≈	2.3669E+11(1.1592E+11)
F14	7.2795E+11(1.8277E+11) +	4.5581E+11(9.6660E+10)
F15	2.8232E+09(2.5522E+09) +	8.8332E+07(2.1918E+07)
+/-	4/1/0	

construction. LSEO-S3-R, which randomly selects a subset of variables to construct the subproblem, may fail to capture the critical variables that significantly influence the objective function. Consequently, LSEO-S3-R suffers from the “curse of dimensionality” in a different form—instead of struggling with the full high-dimensional space, it grapples with a potentially irrelevant or suboptimal subspace. In contrast, the surrogate-assisted subproblem selection strategy of LSEO-S3 intelligently identifies and focuses on the most influential dimensions. This targeted dimension reduction allows the algorithm to optimize over a more relevant subspace, leading to a more effective exploration of the solution landscape and, ultimately, superior performance in solving large-scale expensive optimization problems.

To further illustrate the performance difference between the two methods, we present their convergence trends in Fig. 2. The graphs clearly show that LSEO-S3 exhibits superior efficiency in terms of objective function evaluations. Specifically, LSEO-S3 consistently requires fewer evaluations to achieve comparable or better objective values across the test problems. This accelerated convergence highlights the enhanced capability of LSEO-S3 in navigating the complex landscape of large-scale expensive optimization problems.

The faster convergence of LSEO-S3 can be attributed to its exploration of the solution space with the help of the surrogate-assisted dimension reduction strategy. By focusing on the most influential dimensions, LSEO-S3 makes more effective use of each expensive function evaluation, leading to a fast convergence to the promising regions in the search space.

C. Effectiveness of the Modified SL-PSO

To assess the efficacy of our proposed velocity updating strategy, we conduct a comparative analysis between LSEO-S3 and a variant we term LSEO-S3-O. This variant retains all other aspects of LSEO-S3 but replaces our modified velocity updating equation with the canonical version from SL-PSO. We evaluate both LSEO-S3 and LSEO-S3-O on a diverse set of test problems: F1, F7, F11, F14, and F15.

Table II shows the statistical results obtained by LSEO-S3-O and LSEO-S3. According to Table II, we can observe that LSEO-S3 achieves better results in all cases. This superior performance can be attributed to the key modification in the third component of (7). Specifically, our approach directs individuals to learn from the best solution found so far, rather than from the average position of all individuals in the population. This modification appears to enhance the exploration and exploitation balance of LSEO-S3. Moreover, the strategy of updating only a small subset of dimensions for each solution at each iteration benefits large-scale optimization. By learning from the best solution found so far within this reduced dimensional space, the algorithm maintains a strong exploitation ability without easily falling into local optima. This careful balance allows LSEO-S3 to navigate the complex landscape of large-scale expensive problems more effectively.

It should be noted that only one iteration is carried out for the modified SL-PSO at each generation, i.e., $n_{iter} = 1$. To investigate the impact of the parameter n_{iter} , we conduct a

TABLE II
MEDIAN AND MAD OBTAINED BY LSEO-S3-O AND LSEO-S3 ON F1, F7, F11, F14, F15 TEST PROBLEMS. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED

Prob.	LSEO-S3-O	LSEO-S3
F1	8.0636E+10(4.4496E+09) +	5.0344E+09(4.6611E+08)
F7	9.5285E+11(3.9040E+11) +	1.8950E+09(4.4321E+08)
F11	1.0167E+14(2.9017E+13) +	2.3669E+11(1.1592E+11)
F14	1.1403E+14(2.9612E+13) +	4.5581E+11(9.6660E+10)
F15	3.8953E+14(1.1689E+14) +	8.8332E+07(2.1918E+07)
+/-	5/0/0	

number of experiments on LSEO-S3 by setting n_{lter} to 1, 2, 5, and 10, respectively. Table III shows the experimental results for various values of parameter n_{lter} on five test problems. From Table III, we can see that LSEO-S3 achieves best results when n_{lter} is set to 1 and the performance of LSEOP is decreased with the increase of generations for the subproblem optimization. Ideally, the optimal results will be better if the optimization of subproblem is conducted on more generations. However, although the selected dimensions for the subproblem in Algorithm 3 are identified as the most important dimensions for LSEOP, they are not able to fully represent the large-scale problem. Thus, more generations provided for subproblem optimization may result in the population deviating from the correct search direction of LSEOP because the model for subproblem is not able to fully represent the model for the large-scale expensive problem. Therefore, in our method, only one generation is used for optimization of subproblem.

D. Comparative Experiments on CEC'2013

To the best of our knowledge, the field of SAEAs for expensive LSOPs remains relatively unexplored. To evaluate the effectiveness of LSEO-S3 in this context, we conduct a comparative study against several state-of-the-art optimization algorithms. The algorithms selected for comparison are:

- 1) *SEA-LEEER-G* [52]: The algorithm employs a random grouping technique to partition the LSOP into several low-dimensional subproblems. For each subproblem, an ensemble model including a global and a local model is trained to provide fitness predictions. SEA-LEEER-G performs quite well on large-scale expensive optimization problems.
- 2) *SAEA-RFS* [51]: This algorithm employs a random feature selection technique to choose decision variables from the original LSOP and create a set of subproblems at each generation. Each subproblem is sequentially optimized with the support of its corresponding surrogate model. SAEA-RFS also showcases good performance in solving large-scale expensive optimization problems.
- 3) *SADE-AMSS* [53]: This algorithm utilizes PCA and random decision variable selection to construct multiple subspaces. It then performs adaptive search within these constructed subspaces using three search strategies. SADE-AMSS demonstrates excellent performance in solving large-scale expensive optimization problems.

- 4) *L2SMEA* [54]: This algorithm establishes multiple linear subspaces based on a set of elite solutions. This approach enables effective proxy modeling in the original high-dimensional space by building models on these lower-dimensional subspaces. L2SMEA demonstrates good performance on the CEC'2010 benchmark suite, showcasing its effectiveness in handling LSOPs. However, when tested against the CEC'2013 problem set, the performance of L2SMEA is less impressive.

We evaluated the performance of LSEO-S3 against SEA-LEEER-G, SAEA-RFS, SADE-AMSS, and L2SMEA using 15 CEC'2013 test instances. To ensure fair comparison, all algorithms used parameter settings as recommended in their original publications. The statistical results in terms of the median and MAD values of the optimal objective values achieved by all algorithms under comparison are summarized in Table IV.

According to Table IV, the LSEO-S3 algorithm shows superior performance on CEC'2013 benchmark problems. It achieves the best (highlighted) results in 10 out of 15 test functions (F1, F4, F6, F7, F8, F11, F12, F13, F14, F15), significantly outperforming the other algorithms. L2SMEA shows the second-best performance, obtaining optimal results in 4 functions (F3, F5, F9, F10). SEA-LEEER-G achieves the best result in only one function (F2). Notably, SAEA-RFS and SADE-AMSS do not achieve the optimal result in any of the test functions.

SEA-LEEER-G shows a strong performance in tackling fully separable expensive LSOP problems, e.g., F1, F2, and F3. A possible explanation is that its randomized grouping strategy can enhance the probability of assigning correlated variables to the same group. However, it leads to poorer performance on partially separable and inseparable problems. Similarly, SAEA-RFS showcases good performance on fully separable problems, thanks to its random feature selection strategy. SADE-AMSS exhibits strong performance across a range of problem types: fully separable (F2, F3, F15) and partially separable (F5, F9). The main advantage of SADE-AMSS comes from the use of multiple subspace surrogate modeling and three adaptive search strategies to escape local optima. L2SMEA demonstrates excellent performance across a range of function types, including fully separable (F2), partially additive separable with a separable subcomponent (F5), partially additive separable without a separable subcomponent (F9), and overlapping functions (F14). This superior performance can be attributed to the relative simplicity of these functions within their respective categories. As a result, L2SMEA achieves better results on the CEC'2010 benchmark compared to the more complex CEC'2013 benchmark. However, as the optimization progresses, the tendency of elite solutions to cluster together makes it increasingly difficult to generate diverse linear subspaces. This clustering limits the ability of L2SMEA to effectively explore and cover the entire solution landscape.

Our proposed method presents better or competitive performance on partially additive separable functions with a separable subcomponent (F4-F7) except for F5, partially additive separable functions with no separable subcomponents

TABLE III

STATISTICAL RESULTS OF THE OPTIMAL OBJECTIVE VALUES, GIVEN IN MEDIAN AND MAD, ON F1, F7, F11, F14, AND F15 PROBLEMS OBTAINED BY LSEO-S3 WITH DIFFERENT PARAMETER $nIter$. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED

Prob.	$nIter = 1$	rank	$nIter = 2$	rank	$nIter = 5$	rank	$nIter = 10$	rank
F1	5.0344E+09(4.6611E+08)	1	7.4561E+09(1.3119E+09)	2	1.2822E+10(2.9002E+09)	3	1.8747E+10(5.6360E+09)	4
F7	1.8950E+09(4.4321E+08)	1	2.8611E+09(6.3996E+08)	2	3.0249E+09(5.9485E+08)	3	3.5397E+09(1.7306E+09)	4
F11	2.3669E+11(1.1592E+11)	1	2.3726E+11(4.2116E+10)	2	2.9765E+11(1.2079E+11)	3	5.7938E+11(1.6001E+11)	4
F14	4.5581E+11(9.6660E+10)	2	4.2469E+11(1.0929E+11)	1	6.3661E+11(2.2041E+11)	3	7.4345E+11(1.8024E+11)	4
F15	8.8332E+07(2.1918E+07)	1	4.3579E+08(3.5152E+08)	2	6.5891E+11(6.1554E+11)	3	8.2577E+12(7.1431E+12)	4
Average sort		1.2		1.8		3		4

TABLE IV

STATISTICAL RESULTS OF THE OPTIMAL FITNESS VALUES OBTAINED BY DIFFERENT ALGORITHMS ON THE CEC' 2013 BENCHMARK PROBLEMS ARE PRESENTED IN TERMS OF MEDIAN AND MAD. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED

Prob.	SEA-LEEA-RG	SAEA-RFS	SADE-AMSS	L2SMEA	LSEO-S3
F1	5.0580E+09(4.1343E+08) $\approx$	7.1040E+09(5.5284E+08) +	8.1165E+09(3.8075E+08) +	1.0159E+10(3.2276E+08) +	5.0344E+09(4.6611E+08)
F2	1.7130E+04(3.9444E+02) -	1.9990E+04(4.7842E+02) -	2.2632E+04(3.5924E+02) -	2.3679E+04(6.7559E+02) -	2.8082E+04(6.3750E+02)
F3	2.0780E+01(1.7927E-02) -	2.0920E+01(1.1895E-02) -	2.1002E+01(1.3550E-02) -	2.1629E+01(2.5223E-03) +	2.1221E+01(9.1000E-03)
F4	1.1570E+12(2.7489E+11) +	1.1980E+12(2.9443E+11) +	5.4928E+11(1.5849E+11) +	1.0757E+12(8.7632E+10) +	3.1863E+11(4.2839E+10)
F5	2.6160E+07(3.2960E+06) +	2.3940E+07(2.3474E+06) +	1.8561E+07(1.8437E+06) $\approx$	3.9745E+06(3.4863E+05) -	1.7075E+07(1.9882E+06)
F6	1.0600E+06(2.3701E+03) $\approx$	1.0610E+06(1.7974E+03) +	1.0627E+06(1.4284E+03) +	1.0663E+06(7.9395E+02) +	1.0582E+06(2.2800E+03)
F7	7.2250E+09(1.9855E+09) +	6.9320E+09(2.1430E+09) +	5.5998E+09(1.5465E+09) +	6.9737E+10(1.4444E+10) +	1.8950E+09(4.4321E+08)
F8	3.8800E+16(1.3258E+16) +	1.9640E+16(6.4764E+15) +	1.7871E+16(5.0028E+15) +	1.1752E+16(3.2257E+15) $\approx$	8.6994E+15(2.1620E+15)
F9	1.7500E+09(2.5236E+08) +	1.6060E+09(2.3199E+08) +	1.3271E+09(1.0309E+08) -	5.0352E+08(2.0654E+07) -	1.4707E+09(1.9394E+08)
F10	9.4210E+07(2.8081E+03) $\approx$	9.4550E+07(4.2776E+05) +	9.4496E+07(3.5725E+05) +	9.5227E+07(1.8861E+05) +	9.3848E+07(5.5030E+05)
F11	8.4290E+11(2.4339E+11) +	8.8610E+11(3.0049E+11) +	5.8184E+11(2.1459E+11) +	3.6925E+11(3.6362E+10) +	2.3669E+11(1.1592E+11)
F12	1.0110E+12(1.3607E+10) +	4.8910E+11(2.8418E+10) +	1.1677E+11(1.1082E+10) +	1.1124E+11(1.1745E+10) +	5.9552E+10(7.1232E+09)
F13	8.2660E+10(1.3707E+10) +	1.0890E+11(1.3510E+10) +	9.3184E+10(1.0359E+10) +	1.2465E+11(2.2637E+10) +	3.2256E+10(3.2393E+09)
F14	1.1280E+12(1.7040E+11) +	1.1210E+12(1.1850E+11) +	1.0919E+12(2.6952E+11) +	2.4569E+11(3.9220E+10) -	4.5581E+11(9.6660E+10)
F15	1.8490E+08(4.2987E+07) +	3.1400E+08(8.1643E+07) +	1.1637E+08(2.3121E+07) $\approx$	3.8976E+11(9.7694E+10) +	8.8332E+07(2.1918E+07)
+/ $\approx$ /-	10/3/2	13/0/2	10/2/3	11/0/4	

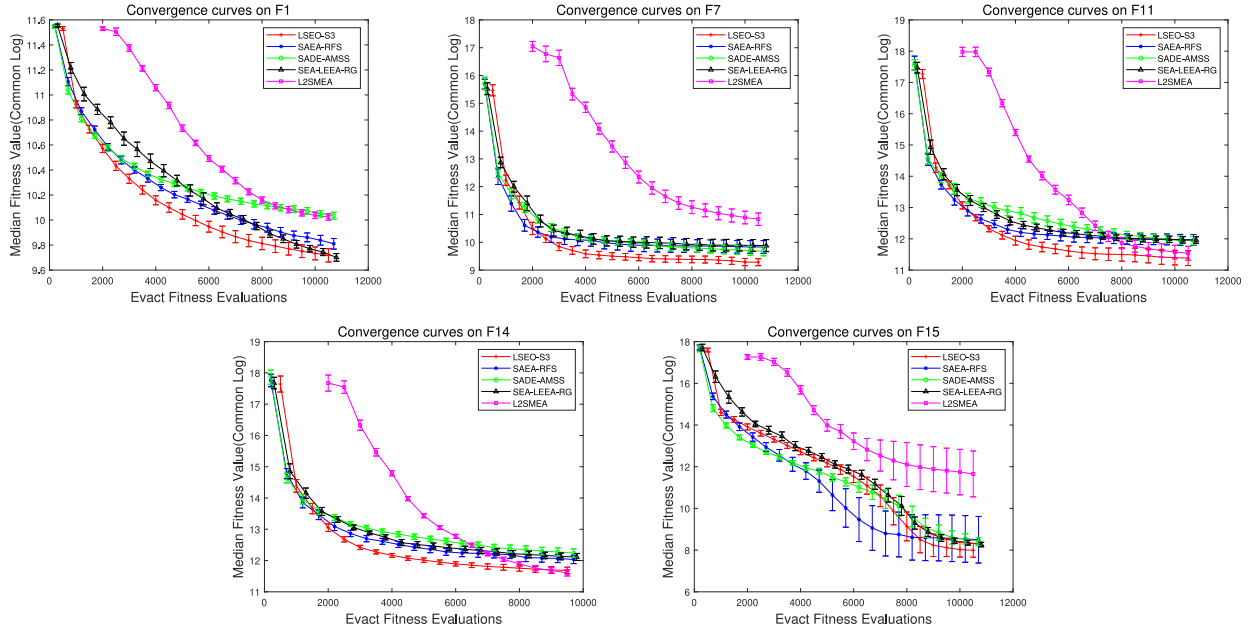


Fig. 3. Convergence of median value on the optimal solution found so far obtained by LSEO-S3 and other four algorithms on F1, F7, F11, F14, and F15.

(F8–F11) except for F9, overlapping functions (F12–F14) except for F14, and nonseparable function (F15). Regarding fully separable functions, it outperforms other methods on F1 but struggles with F2 and F3. In fully separable functions, variables can be optimized independently to find the global optimum. Consequently, our approach that utilizing the proposed surrogate assisted subproblem selection may have a limited impact on the final optimization results for these specific cases. For the partially separable problems (F4–F14), LSEO-S3 shows superior performance compared to SEA-LEEA-RG,

SAEA-RFS, SADE-AMSS, and L2SMEA in most cases. However, LSEO-S3 exhibits inferior performance on F5, F9, and F14 compared to L2SMEA. The main reason is that the linear subspace surrogate modeling in L2SMEA is able to learn the landscape efficiently. For the completely nonseparable problem (F15), LSEO-S3 gets superior performance compared to all algorithms under comparison, validating the effectiveness of LSEO-S3 in handling complex, nonseparable optimization landscapes. Furthermore, this observation points out the potential of our method in tackling challenging optimization

TABLE V

STATISTICAL RESULTS (MEDIAN AND MAD) OF SEA-LEEA-RG, SAEA-RFS, SADE-AMSS, L2SMEA AND THE PROPOSED LSEO-S3 ON F1, F4, F11, F15 AND F20 OF CEC'2010 BENCHMARK PROBLEMS WITH 1000 AND 2000 DIMENSIONS. THE BEST RESULT IN EACH ROW IS HIGHLIGHTED

Prob.	D	SEA-LEEA-RG	SAEA-RFS	SADE-AMSS	L2SMEA	LSEO-S3
F1	1000	5.4573E+09(4.4761E+08) $\approx$	1.5874E+10(1.6035E+09) +	1.4162E+10(6.7144E+08) +	9.3313E+09(3.6862E+08) +	5.1864E+09(5.6439E+08)
	2000	4.2237E+10(1.8965E+09) +	8.9383E+10(5.5988E+09) +	4.1973E+10(1.7061E+09) +	1.2547E+11(5.7733E+09) +	2.1375E+10(1.3532E+09)
F4	1000	7.3124E+13(1.3710E+13) +	8.7143E+13(2.2042E+13) +	6.1178E+13(1.5695E+13) +	3.2402E+13(8.0718E+12) +	2.0973E+13(3.6126E+12)
	2000	9.8611E+13(3.0032E+13) +	8.5907E+13(1.9472E+13) +	4.7841E+13(8.6252E+12) +	6.1526E+14(1.5744E+14) +	2.3574E+13(3.1913E+12)
F11	1000	2.2756E+02(4.1745E-01) -	2.3655E+02(3.7430E-01) +	2.3486E+02(2.5518E-01) +	2.1207E+02(4.6872E+00) -	2.3318E+02(3.0600E-01)
	2000	4.4155E+02(5.1771E-01) -	4.5139E+02(5.9660E-01) +	4.4905E+02(8.0454E-01) +	4.3634E+02(3.5357E+00) -	4.4670E+02(3.7800E-01)
F15	1000	1.8875E+04(2.8339E+02) +	2.1076E+04(4.6086E+02) +	2.0412E+04(3.6331E+02) +	1.2534E+04(5.0203E+02) -	1.4063E+04(3.2820E+02)
	2000	4.1392E+04(3.8639E+02) +	4.5984E+04(9.1138E+02) +	4.1693E+04(1.0321E+03) +	3.8419E+04(6.0772E+02) +	3.2818E+04(4.9210E+02)
F20	1000	3.5545E+11(2.6281E+10) +	1.1617E+12(7.0611E+10) +	7.6588E+11(4.2680E+10) +	1.1000E+11(1.6191E+10) +	5.9828E+10(7.4134E+09)
	2000	2.3388E+12(6.6989E+10) +	5.0804E+12(2.4402E+11) +	9.3286E+11(2.6784E+10) +	4.0663E+12(1.9406E+11) +	5.7761E+11(2.8590E+10)
+/ $\approx$ /-		7/1/2	10/0/0	10/0/0	7/0/3	

problems, particularly those with high degrees of variable interdependence.

To provide a more detailed analysis of the optimization performance across all algorithms, we have plotted convergence curves on various functions in Fig. 3. The convergence curves are plotted on a coordinate system where the X-axis represents the number of function evaluations, while the Y-axis displays the mean fitness value achieved. As illustrated in Fig. 3, the proposed LSEO-S3 algorithm consistently shows superior or highly competitive performance across all tested functions compared to the other algorithms under evaluation. The convergence curves reveal that LSEO-S3 not only achieves better final fitness values in most cases but also exhibits faster convergence rates. Specifically, LSEO-S3 clearly outperforms its competitors throughout the optimization process on functions F1, F7, and F11. Even in cases where the performance gap is narrower, such as in F14 and F15, LSEO-S3 matches the best-performing alternatives. These results justify that the proposed algorithm offers an efficient solution to solving a diverse range of optimization problems, from fully separable to nonseparable large scale optimization functions. The results suggest that the algorithm is effective in addressing a variety of LSOPs.

E. Comparative Experiments on CEC'2010

In section IV-D, the proposed algorithm has shown competitive performance in solving CEC'2013 test instances. To provide a more comprehensive evaluation, we extended our comparison to five representative functions selected from the CEC'2010 benchmark suite. The benchmark problems in CEC'2010 can be categorized into three groups, fully separable problems (F1–F3), partially separable problems (F4–F18), and nonseparable problems (F19–F20). These problems also vary in their landscape characteristics: F1, F4, F7, F9, F12, F14, F17, and F19 feature single-peak landscapes, while the others present symmetric multi-peak landscapes. To ensure a diverse and representative test set, we selected problems F1, F4, F11, F15, and F20 for our comparative study. To further test the scalability and effectiveness of these algorithms, the dimension of the selected test instances is set to 1000 and 2000, respectively. Table V presents the statistical results of the median and MAD of the best objective values achieved by each algorithm on F1, F4, F11, F15, and F20 from CEC'2010, respectively.

The results in Table V reveal that the proposed algorithm exhibited remarkable performance, particularly when tackling high-dimensional problems. Specifically, SEA-LEEA-RG, SAEA-RFS, and SADE-AMSS consistently achieve worse performance than LSEO-S3 across all tested problems, failing to achieve optimal results in any instance. Interestingly, L2SMEA shows promising performance on 1000- and 2000-D F11 and 1000-D F15.

Our experimental results on 15 test instances from CEC'2013 and additional high-dimensional problems from CEC'2010 reveal the strengths of our proposed algorithm. Our method achieves promising optimization performance on most test instances, and maintains its competitive performance when tested on higher-dimensional problems. These observations highlight that while no single algorithm can effectively solve all problem types, our approach offers significant advantages in tackling a wide range of complex optimization tasks, especially those involving partial separability or nonseparability, and scales well to higher dimensions.

V. CONCLUSION AND FUTURE WORK

In this article, an efficient large-scale expensive optimization algorithm via surrogate-assisted sub-problem selection is proposed. We employ surrogate models to assist in dimension reduction, enhancing the probability of identifying and selecting critical dimensions. This strategy accelerates the search for optimal solutions while enabling more efficient offspring generation in the selected dimensions. We further enhance the algorithm by modifying the velocity update mechanism in SL-PSO, carefully balancing exploration capabilities with expedited search toward promising directions. Experimental results reveal the superiority of our proposed algorithm compared with four state-of-the-art algorithms, with particularly notable performance on partially separable and nonseparable problems.

Dimension reduction plays a critical role in addressing large-scale expensive optimization problems, yet it inevitably entails some loss of information from the original problem formulation. Therefore, a possible future direction is to more effectively identify and select influential dimensions while minimizing information loss.

REFERENCES

- [1] S. Nguyen, M. Zhang, M. Johnston, and K. C. Tan, "Automatic programming via iterated local search for dynamic job shop scheduling," *IEEE Trans. Cybern.*, vol. 45, no. 1, pp. 1–14, Jan. 2015.

- [2] K. Deb, A. R. Reddy, and G. Singh, "Optimal scheduling of casting sequence using genetic algorithms," *Mater. Manuf. Process.*, vol. 3, no. 3, pp. 409–432, 2003.
- [3] K. Deb and C. Myburgh, "Breaking the billion-variable barrier in real-world optimization using a customized evolutionary algorithm," in *Proc. Genetic Evol. Comput. Conf.*, 2016, pp. 653–660.
- [4] M. Xiao, J. Zhang, K. Cai, X. Cao, and K. Tang, "Cooperative co-evolution with weighted random grouping for large-scale crossing waypoints locating in air route network," in *Proc. IEEE 23rd Int. Conf. Tools Artif. Intell.*, 2011, pp. 215–222.
- [5] S. Mahdavi, M. E. Shiri, and S. Rahnamayan, "Metaheuristics in large-scale global continues optimization: A survey," *Inf. Sci.*, vol. 295, pp. 407–428, Feb. 2015.
- [6] P. Jiang, Y. Cheng, and J. Liu, "Cooperative Bayesian optimization with hybrid grouping strategy and sample transfer for expensive large-scale black-box problems," *Knowl.-Based Syst.*, vol. 254, Oct. 2022, Art. no. 109633.
- [7] A. Van Den Bos, "Complex gradient and Hessian," *IEE Proc.-Vis., Image Signal Process.*, vol. 141, no. 6, pp. 380–382, 1994.
- [8] J. Liu, R. Sarker, S. Elsayed, D. Essam, and N. Siswanto, "Large-scale evolutionary optimization: A review and comparative study," *Swarm Evol. Comput.*, vol. 85, Mar. 2024, Art. no. 101466.
- [9] Y. Wu, N. Yang, L. Chen, Y. Tian, and Z. Tang, "Directed quick search guided evolutionary framework for large-scale multi-objective optimization problems," *Expert Syst. Appl.*, vol. 239, Apr. 2024, Art. no. 122370.
- [10] J. Jian, Z. Zhan, and J. Zhang, "Large-scale evolutionary optimization: A survey and experimental comparative study," *Int. J. Mach. Learn. Cybern.*, vol. 11, no. 3, pp. 729–745, 2020.
- [11] Y. Tian et al., "Evolutionary large-scale multi-objective optimization: A survey," *ACM Comput. Surv.*, vol. 54, no. 8, pp. 1–34, 2021.
- [12] M. A. Potter and K. A. De Jong, "A cooperative co-evolutionary approach to function," in *Proc. Int. Conf. Parallel Problem Solving Nat.*, 1994, pp. 249–257.
- [13] Z. Yang, K. Tang, and X. Yao, "Large scale evolutionary optimization using cooperative coevolution," *Inf. Sci.*, vol. 178, no. 15, pp. 2985–2999, 2008.
- [14] W. Chen, T. Weise, Z. Yang, and K. Tang, "Large-scale global optimization using cooperative coevolution with variable interaction learning," in *Proc. 11th Int. Conf. Parallel Problem Solving Nat.*, 2010, pp. 300–309.
- [15] X. Li and X. Yao, "Cooperatively coevolving particle swarms for large scale optimization," *IEEE Trans. Evol. Comput.*, vol. 16, no. 2, pp. 210–224, Apr. 2012.
- [16] Y. Sun, M. Kirley, and S. K. Halgamuge, "Extended differential grouping for large scale global optimization with direct and indirect variable interactions," in *Proc. Annu. Conf. Genetic Evol. Comput.*, 2015, pp. 313–320.
- [17] Y. Mei, M. N. Omidvar, X. Li, and X. Yao, "A competitive divide-and-conquer algorithm for unconstrained large-scale black-box optimization," *ACM Trans. Math. Softw.*, vol. 42, no. 2, pp. 1–24, 2016.
- [18] M. N. Omidvar, M. Yang, Y. Mei, X. Li, and X. Yao, "DG2: A faster and more accurate differential grouping for large-scale black-box optimization," *IEEE Trans. Evol. Comput.*, vol. 21, no. 6, pp. 929–942, 2017.
- [19] M. Yang et al., "Efficient resource allocation in cooperative co-evolution for large-scale global optimization," *IEEE Trans. Evol. Comput.*, vol. 21, no. 4, pp. 493–505, Aug. 2017.
- [20] M. Yang, A. Zhou, C. Li, J. Guan, and X. Yan, "CCFR2: A more efficient cooperative co-evolutionary framework for large-scale global optimization," *Inf. Sci.*, vol. 512, pp. 64–79, Feb. 2020.
- [21] R. Liu, R. Ren, J. Liu, and J. Liu, "A clustering and dimensionality reduction based evolutionary algorithm for large-scale multi-objective problems," *Appl. Soft Comput.*, vol. 89, Apr. 2020, Art. no. 106120.
- [22] C. He et al., "Accelerating large-scale multiobjective optimization via problem reformulation," *IEEE Trans. Evol. Comput.*, vol. 23, no. 6, pp. 949–961, Dec. 2019.
- [23] Y. Tian, C. Lu, X. Zhang, K. C. Tan, and Y. Jin, "Solving large-scale multiobjective optimization problems with sparse optimal solutions via unsupervised neural networks," *IEEE Trans. Cybern.*, vol. 51, no. 6, pp. 3115–3128, Jun. 2021.
- [24] Q. Yang et al., "Random contrastive interaction for particle swarm optimization in high-dimensional environment," *IEEE Trans. Evol. Comput.*, vol. 28, no. 4, pp. 933–949, Aug. 2024.
- [25] R. Cheng and Y. Jin, "A competitive swarm optimizer for large scale optimization," *IEEE Trans. Cybern.*, vol. 45, no. 2, pp. 191–204, Feb. 2015.
- [26] H. Wang, M. Liang, C. Sun, G. Zhang, and L. Xie, "Multiple-strategy learning particle swarm optimization for large-scale optimization problems," *Complex Intell. Syst.*, vol. 7, pp. 1–16, Feb. 2021.
- [27] R. Cheng and Y. Jin, "A social learning particle swarm optimization algorithm for scalable optimization," *Inf. Sci.*, vol. 291, pp. 43–60, Jan. 2015.
- [28] Z.-J. Wang, J.-R. Jian, Z.-H. Zhan, Y. Li, S. Kwong, and J. Zhang, "Gene targeting differential evolution: A simple and efficient method for large-scale optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 4, pp. 964–979, Aug. 2023.
- [29] D.-H. Kim, D.-H. Choi, and H.-S. Kim, "Design optimization of a carbon fiber reinforced composite automotive lower arm," *Composites Part B, Eng.*, vol. 58, pp. 400–407, Mar. 2014.
- [30] H. Khayyam, M. Naebe, O. Zabihi, R. Zamani, S. Atkiss, and B. Fox, "Dynamic prediction models and optimization of polyacrylonitrile (PAN) stabilization processes for production of carbon fiber," *IEEE Trans. Ind. Informat.*, vol. 11, no. 4, pp. 887–896, Aug. 2015.
- [31] Y. Jin, "Surrogate-assisted evolutionary computation: Recent advances and future challenges," *Swarm Evol. Comput.*, vol. 1, no. 2, pp. 61–70, 2011.
- [32] Y. Jin, M. Olhofer, and B. Sendhoff, "A framework for evolutionary optimization with approximate fitness functions," *IEEE Trans. Evol. Comput.*, vol. 6, no. 5, pp. 481–494, Oct. 2002.
- [33] H. Wang, Y. Jin, C. Sun, and J. Doherty, "Offline data-driven evolutionary optimization using selective surrogate ensembles," *IEEE Trans. Evol. Comput.*, vol. 23, no. 2, pp. 203–216, Apr. 2019.
- [34] X. Y. Sun, D. W. Gong, and X. P. Ma, "Directed fuzzy graph-based surrogate model-assisted interactive genetic algorithms with uncertain individual's fitness," in *Proc. IEEE Congr. Evol. Comput.*, 2009, pp. 2395–2402.
- [35] R. Mallipeddi and M. Lee, "Surrogate model assisted ensemble differential evolution algorithm," in *Proc. IEEE Congr. Evol. Comput.*, 2012, pp. 1–8.
- [36] R. G. Regis, "Particle swarm with radial basis function surrogates for expensive black-box optimization," *J. Comput. Sci.*, vol. 5, no. 1, pp. 12–23, 2014.
- [37] Y. Jin, "A comprehensive survey of fitness approximation in evolutionary computation," *Soft Comput.*, vol. 9, no. 1, pp. 3–12, 2005.
- [38] Y. Lian and M.-S. Liou, "Multiobjective optimization using coupled response surface model and evolutionary algorithm," *AIAA J.*, vol. 43, no. 6, pp. 1316–1325, 2005.
- [39] J. Tian, Y. Tan, J. Zeng, C. Sun, and Y. Jin, "Multiobjective infill criterion driven Gaussian process-assisted particle swarm optimization of high-dimensional expensive problems," *IEEE Trans. Evol. Comput.*, vol. 23, no. 3, pp. 459–472, Jun. 2019.
- [40] H. Yu, Y. Tan, J. Zeng, C. Sun, and Y. Jin, "Surrogate-assisted hierarchical particle swarm optimization," *Inf. Sci.*, vols. 454–455, pp. 59–72, Jul. 2018.
- [41] X. Cheng, Y. Yu, and W. Hu, "Multi-surrogate-assisted stochastic fractal search algorithm for high-dimensional expensive problems," *Inf. Sci.*, vol. 640, Sep. 2023, Art. no. 119035.
- [42] H. Yu, Y. Tan, C. Sun, and J. Zeng, "A generation-based optimal restart strategy for surrogate-assisted social learning particle swarm optimization," *Knowl.-Based Syst.*, vol. 163, pp. 14–25, Jan. 2019.
- [43] X. Wang, G. G. Wang, B. Song, P. Wang, and Y. Wang, "A novel evolutionary sampling assisted optimization method for high-dimensional expensive problems," *IEEE Trans. Evol. Comput.*, vol. 23, no. 5, pp. 815–827, Oct. 2019.
- [44] C. Sun, Y. Jin, R. Cheng, J. Ding, and J. Zeng, "Surrogate-assisted cooperative swarm optimization of high-dimensional expensive problems," *IEEE Trans. Evol. Comput.*, vol. 21, no. 4, pp. 644–660, Aug. 2017.
- [45] I. De Falco, A. Della Cioppa, and G. A. Trunfio, "Investigating surrogate-assisted cooperative coevolution for large-scale global optimization," *Inf. Sci.*, vol. 482, pp. 1–26, May 2019.
- [46] I. De Falco, A. D. Cioppa, and G. A. Trunfio, "Large scale optimization of computationally expensive functions: An approach based on parallel cooperative coevolution and fitness metamodeling," in *Proc. Genetic Evol. Comput. Conf. Companion*, 2017, pp. 1788–1795.
- [47] M. A. Potter and K. A. De Jong, "A cooperative coevolutionary approach to function optimization," in *Proc. Int. Conf. Parallel Problem Solving Nat.*, 1994, pp. 249–257.
- [48] M. N. Omidvar, X. Li, Z. Yang, and X. Yao, "Cooperative co-evolution for large scale optimization through more frequent random grouping," in *Proc. IEEE Congr. Evol. Comput.*, 2010, pp. 1–8.

- [49] M. N. Omidvar, X. Li, Y. Mei, and X. Yao, "Cooperative co-evolution with differential grouping for large scale optimization," *IEEE Trans. Evol. Comput.*, vol. 18, no. 3, pp. 378–393, Jun. 2014.
- [50] F. Li, X. Cai, L. Gao, and W. Shen, "A surrogate-assisted multiswarm optimization algorithm for high-dimensional computationally expensive problems," *IEEE Trans. Cybern.*, vol. 51, no. 3, pp. 1390–1402, Mar. 2021.
- [51] G. Fu, C. Sun, Y. Tan, G. Zhang, and Y. Jin, "A surrogate-assisted evolutionary algorithm with random feature selection for large-scale expensive problems," in *Proc. Int. Conf. Parallel Problem Solving Nat.*, 2020, pp. 125–139.
- [52] M. Sun, C. Sun, X. Li, G. Zhang, and F. Akhtar, "Surrogate ensemble assisted large-scale expensive optimization with random grouping," *Inf. Sci.*, vol. 615, pp. 226–237, Nov. 2022.
- [53] H. Gu, H. Wang, and Y. Jin, "Surrogate-assisted differential evolution with adaptive multi-subspace search for large-scale expensive optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 6, pp. 1765–1779, Dec. 2023.
- [54] L. Si, X. Zhang, Y. Tian, S. Yang, L. Zhang, and Y. Jin, "Linear subspace surrogate modeling for large-scale expensive single/multi-objective optimization," *IEEE Trans. Evol. Comput.*, early access, Sep. 27, 2023, doi: [10.1109/TEVC.2023.3319640](https://doi.org/10.1109/TEVC.2023.3319640).
- [55] M. J. Er, S. Wu, J. Lu, and H. L. Toh, "Face recognition with radial basis function (RBF) neural networks," *IEEE Trans. Neural Netw.*, vol. 13, no. 3, pp. 697–710, May 2002.
- [56] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *J. Mach. Learn. Res.*, vol. 13, no. 2, pp. 281–305, 2012.
- [57] X. Li, K. Tang, M. N. Omidvar, Z. Yang, K. Qin, and H. China, "Benchmark functions for the CEC 2013 special session and competition on large-scale global optimization," *Gene*, vol. 7, no. 33, p. 8, 2013.
- [58] R. G. Regis and C. A. Shoemaker, "Combining radial basis function surrogates and dynamic coordinate search in high-dimensional expensive black-box optimization," *Eng. Optim.*, vol. 45, no. 5, pp. 1–27, 2012.
- [59] X. Cai, L. Gao, X. Li, and H. Qiu, "Surrogate-guided differential evolution algorithm for high dimensional expensive problems," *Swarm Evol. Comput.*, vol. 48, pp. 288–311, Aug. 2019.
- [60] J. Derrac, S. García, D. Molina, and F. Herrera, "A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms," *Swarm Evol. Comput.*, vol. 1, no. 1, pp. 3–18, 2011.



Kaili Zhao received the B.Sc. degree from the Taiyuan University of Science and Technology, Taiyuan, China, in 2013, and the M.Sc. degree from Jilin University, Changchun, China, in 2016. She is currently pursuing the Ph.D. degree with the School of Electronic Information Engineering, Taiyuan University of Science and Technology.

Her research interests include large-scale expensive optimization, surrogate-assisted evolutionary optimization, and their applications to practical engineering problems.



Xilu Wang (Associate Member, IEEE) received the B.Sc. degree from the Harbin Institute of Technology, Harbin, China, in 2016, the M.Sc. degree from Xidian University, Xi'an, China, in 2018, and the Ph.D. degree from University of Surrey, Surrey, U.K., in 2022.

She is currently a Surrey Future Fellow with the School of Computer Science and Electronic Engineering, University of Surrey. Her research interests include multiobjective optimization, surrogate-assisted evolutionary optimization,

Bayesian optimization, fair and federated optimization, and evolutionary machine learning.



Chaoli Sun (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees from Hohai University, Nanjing, China, in 2000 and 2003, respectively, and the Ph.D. degree from the Taiyuan University of Science and Technology, Taiyuan, China, in 2011.

She is a Professor with the School of Computer Science and Technology, Taiyuan University of Science and Technology. Her research interests include swarm intelligence, particle swarm optimization, surrogate-assisted evolutionary optimization, and their applications to practical

engineering problems.

Prof. Sun is an Associate Editor of IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE, *Engineering Applications of Artificial Intelligence*, *Memetic Computing*, and *Complex & Intelligent Systems*. She serves as the Chair for the Intelligent Systems Application Technical Committee of the IEEE Computational Intelligence Society and was the Chair of the Task Force on Data-Driven Evolutionary Optimization of Expensive Problems within the Evolutionary Computation Technical Committee of the IEEE Computational Intelligence Society from 2015 to 2020. She has also served as a Reviewer for journals including IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION, IEEE TRANSACTIONS ON CYBERNETICS, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *IEEE Computational Intelligence Magazine*.



Yaochu Jin (Fellow, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1988, 1991, and 1996, respectively, and the Dr.-Ing. degree from Ruhr University Bochum, Bochum, Germany, in 2001.

He is currently a Chair Professor of Artificial Intelligence with the School of Engineering, Westlake University, Hangzhou. Before joining Westlake University, he was an Alexander von Humboldt Professor for Artificial Intelligence endowed by the German Federal Ministry of Education and Research, Bielefeld University, Bielefeld, Germany, from 2021 to 2023, and a Surrey Distinguished Chair Professor in Computational Intelligence, University of Surrey, Guildford, U.K., from 2010 to 2021. He was a "Finland Distinguished Professor" of University of Jyväskylä, Jyväskylä, Finland, and a "Changjiang Distinguished Visiting Professor," Northeastern University, Shenyang, China, from 2015 to 2017. His main research interests include multiobjective and data-driven evolutionary optimization, evolutionary multiobjective learning, trustworthy AI, and evolutionary developmental AI.

Prof. Jin is the recipient of the 2018, 2021, and 2024 IEEE Transactions on Evolutionary Computation Outstanding Paper Award, and the 2015, 2017, and 2020 IEEE Computational Intelligence Magazine Outstanding Paper Award. He is presently the President of the IEEE Computational Intelligence Society and Editor-in-Chief of *Complex & Intelligent Systems*. He was the Editor-in-Chief of the IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS, an IEEE Distinguished Lecturer from 2013 to 2015 and from 2017 to 2019, and the Vice President for Technical Activities of the IEEE Computational Intelligence Society from 2015 to 2016. He has been named consecutively by the Web of Science as "a Highly Cited Researcher" since 2019. He is a Member of Academia Europaea.



Asad Hayat received the M.S. degree from the Capital University of Science and Technology, Islamabad, Pakistan, in 2021. He is currently pursuing the Ph.D. degree with the school of Electronics Information and Engineering, Taiyuan University of Science and Technology, Taiyuan, China.

His current research interests include evolutionary large-scale optimization, surrogate models, machine learning, and artificial intelligence.